FACULTY OF COMPUTING
UTM Johor Bahru

**SECP3213: BUSINESS INTELLIGENCE**

**2025/2026 – SEMESTER 2**

# PROJECT (30%)

## Part 1: Movie Analytics Using Alteryx Designer

Assume that you are a business analyst, and were asked to analyse the Movie Ticket Sales and Revenue data. You are required to design the workflow in Alteryx Designer to find the answer for the following. In each step, it must have appropriate comments to explain or justify the process conducted in the data analysis.

a) Bring in all the Gross Revenue (GR) data sets and clean them as necessary. Record without Release Date and Distributor info (either null or empty) must be filter out.

b) Combine them and summarize the data so that all the revenue generated by a movie is listed in a single record for that movie.

c) Investigate the resulting data set to see if any movie titles have been used multiple times.

d) Based on the unique record, use the provided look-up table (*Distributor LookUp Table.xlsx*) to replace the truncated name of Distributor with the more complete name provided. Then, save the result in MS Excel (Note: label the file name as *MasterList Movie Records.xlsx*).

e) Calculate the total revenue generated by all movies from 2019-2022 and attach that figure to each row. Then, calculate the distributor percentage accounted for by each movie and order the movie accordingly. You need to prepare two files for the output:

   i) Save the outputs in MS Excel file that contains the following fields only – Movie, Release Date, Distributor, Genre, Percentage of Distributor by Movie. Note – label the file name as *GR by Movie 20192022.xlsx*.
   ii) Generate report (in pdf) to show the top 20 records that obtained highest percentage. Refer **Appendix A** – Sample Report.

f) What is the average number of tickets sold? Which movies sold more than the average number of tickets? Save the output in CSV file. Note – label the file name as *Movie Tickets Above Average. csv*

g) Summarize revenue based on year of release and month of release separately. Save the output in CSV file. Note – label the file name as: (i) *Movie Revenue by Year.csv*; (ii) *Movie Revenue by Month.csv*

**Part 2: Movie Analytics Using Tableau**

Load the dataset *MasterList Movie Records.xlsx* into Tableau.

Your task is to design an interactive dashboard in Tableau that explores key trends and insights from the movie industry based on this dataset. Your dashboard must have the following elements:

- interactive, including filters, drop-downs, or highlighters (e.g., by year or distributor).
- Include proper titles, labels, legends, and tooltips to make your dashboard intuitive and professional.

Answer all business questions below through visualization and provide a brief summary of insights (3–5 bullet points) at the bottom of your dashboard.

1. Which movies generated the highest gross revenue, and how does it compare to the number of tickets sold?
2. What is the trend of total gross revenue over the years?
3. Which distributors dominate the market in terms of gross revenue and number of movies released?
4. What is the seasonal pattern of movie releases and ticket sales?
5. Is there a correlation between release month and box office success?

Note: You may also include other relevant business questions where appropriate.

**Submission – Upload all the file in e-learning:**

1) Full workflow (.**yxmd**) of Part 1.
2) Output file for Part 1: (d), (e-i), (f) and (g)
3) PDF report for Part 1 (e-ii)
4) Tableau Package Workbook **(.twbx**) file of Part 2.

Deadline: **Sunday - 28 June 2026; 11.59 p.m.**

## Appendix A: Sample Report

# Top 20 - Gross Revenue of Movie's Distributor (2019-2022)

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| Movie | Release Date | Distributor | Genre | Percentage of Distributor by Movie |
|---|---|---|---|---|
| Avengers: Endgame | 2019-04-26 | Walt Disney | Action | 4.7 |
| Spider-Man: No Way Home | 2021-12-17 | Sony Pictures | Action | 3.4 |
| The Lion King | 2019-07-19 | Walt Disney | Adventure | 3.0 |
| Frozen II | 2019-11-22 | Walt Disney | Adventure | 2.5 |
| Toy Story 4 | 2019-06-21 | Walt Disney | Adventure | 2.4 |
| Captain Marvel | 2019-03-08 | Walt Disney | Action | 2.4 |
| Star Wars: The Rise of Skyw… | 2019-12-20 | Walt Disney | Adventure | 2.2 |
| Spider-Man: Far From Home | 2019-07-02 | Sony Pictures | Action | 2.2 |
| Aladdin | 2019-05-24 | Walt Disney | Adventure | 2.0 |
| The Batman | 2022-03-04 | Warner Bros. | Action | 1.9 |
| Joker | 2019-10-04 | Warner Bros. | Thriller/Suspense | 1.8 |
| Shang-Chi and the Legend of… | 2021-09-03 | Walt Disney | Action | 1.2 |
| Venom: Let There be Carnage | 2021-10-01 | Sony Pictures | Action | 1.2 |
| It: Chapter Two | 2019-09-06 | Warner Bros. | Horror | 1.2 |
| Bad Boys For Life | 2020-01-17 | Sony Pictures | Action | 1.1 |
| Jumanji: The Next Level | 2019-12-13 | Sony Pictures | Adventure | 1.1 |
| Black Widow | 2021-07-09 | Walt Disney | Action | 1.0 |
| Us | 2019-03-22 | Universal | Thriller/Suspense | 1.0 |
| Fast & Furious Presents: Ho… | 2019-08-02 | Universal | Action | 1.0 |
| F9: The Fast Saga | 2021-06-25 | Universal | Action | 1.0 |